

A Signed Decomposition of Mutual Information via Standardized Residuals

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Abstract

We connect a classical tool from contingency-table analysis; the per-cell standardized residual; to mutual information. For a binary class label and a feature X binned into values $v \in \mathcal{V}$, let $r_v(c)$ denote the standardized residual testing $p(c \mid X=v) = p(c)$. We establish two main results. First, the squared profile $\{r_v(c)^2\}_v$ sums *exactly* to Pearson’s chi-square statistic for the feature–class contingency table; this is an algebraic identity that holds at any sample size. Second, under the null hypothesis of independence and under contiguous alternatives, this same sum is asymptotically equal to $2N$ times the empirical mutual information. We extend the result in two directions: a multi-class generalization, and; of central practical importance for feature selection; a conditional version that decomposes empirical conditional mutual information $\hat{I}(X; C \mid Z)$ into a stratified residual profile, connecting the framework directly to mRMR- and CMIM-style feature selection. The signed structure of the residual profile preserves directionality and admits a hypothesis-testing interpretation, yielding a single object that simultaneously functions as a feature-significance test, a mutual information estimator, and a per-bin attribution diagnostic. We outline an empirical study covering both unconditional and conditional decompositions.

Keywords: Mutual information; standardized residuals; Pearson chi-square; conditional mutual information; feature selection; per-bin attribution; contingency tables.

1 Introduction

Identifying features that are statistically significant with respect to a class label is a foundational task in supervised learning. Two long-standing traditions address this question. The first, rooted in classical statistics, frames feature significance as a hypothesis test; chi-square tests for categorical features, t - or z -tests for numerical ones; yielding a per-feature scalar with an associated p -value. The second, rooted in information theory, measures dependence directly through mutual information $I(X; C)$, which is non-negative, symmetric, and zero exactly when X and C are independent.

Each tradition has trade-offs. Hypothesis tests are interpretable, support per-cell diagnostics through standardized residuals [3, 1], and connect naturally to confidence statements. Mutual

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information is model-free, captures arbitrary nonlinear dependence, and aggregates cleanly across the feature’s domain; but as a scalar, it provides no insight into *which* regions of the feature drive the dependence, nor which class each region favors.

This note observes that the gap between these two traditions is narrower than it appears: a classical object; the per-bin standardized residual; already lives between them, and its profile across the feature’s domain is, in a precise sense, a signed decomposition of mutual information.

For a binary class label $C \in \{c, \bar{c}\}$ and a feature value v , the standardized residual

$$r_v(c) = \sqrt{N_{X_v}} \cdot \frac{\hat{p}(c \mid X=v) - \hat{p}(c)}{\sqrt{\hat{p}(c)(1 - \hat{p}(c))}}$$

is the one-sample z -statistic for the binomial test of $H_0 : p(c \mid X=v) = p(c)$, conditional on the bin count N_{X_v} . Its sign encodes directionality: positive values indicate the bin favors c , negative values indicate the bin favors $\bar{c}$. Standardized residuals of this form are textbook diagnostics in categorical data analysis [3, 1]; the contribution here is not in the statistic but in what its profile reveals when squared and aggregated.

Specifically, we prove that

- (i) $\sum_v r_v(c)^2$ equals Pearson’s chi-square statistic for the feature–class contingency table *exactly* (an algebraic identity, no asymptotics required); and
- (ii) under the null and under contiguous alternatives, this sum equals $2N\hat{I}(X; C) + o_p(1)$, where $\hat{I}$ is the empirical mutual information.

We then extend the result to a conditional decomposition: stratifying by an additional variable Z , the within-slice residual profiles aggregate to a stratified statistic that decomposes empirical conditional mutual information $\hat{I}(X; C \mid Z)$ in the same fashion. This conditional version is what connects the framework to standard MI-based feature selection methods such as mRMR [11] and CMIM [2], which score features by quantities involving conditional mutual information.

Squaring and summing the profile recovers the global (or conditional) mutual information; retaining the signed profile preserves directional, per-bin information that the scalar $\hat{I}$ discards. The result yields a feature-importance framework that is at once a hypothesis-testing object, an MI estimator, and an interpretability tool.

1.1 Contributions

The technical pieces in this paper are individually classical: standardized residuals [3], Pearson’s chi-square test, the chi-square / G-test asymptotic equivalence [5, 1, 6], and Kullback’s identity $G = 2N\hat{I}$ [5]. What we contribute is their assembly into a coherent decomposition framework, together with the conditional extension that makes the framework operational for feature selection:

1. **Exact algebraic identity (Theorem 1, part i).** For binary classification, $\sum_v r_v(c)^2 = \chi^2$. While the underlying contingency-table algebra is elementary, this collapsed form; summing over feature values with the binary class fixed, using a single signed residual per bin; exploits a sign-flip symmetry specific to two-class problems and appears not to be stated explicitly in the literature.
2. **Asymptotic decomposition of mutual information (Theorem 1, part ii).** Under the null and under contiguous alternatives, $\sum_v r_v(c)^2 = 2N\hat{I}(X; C) + o_p(1)$. Each bin contributes share $r_v(c)^2/(2N)$ asymptotically to $\hat{I}(X; C)$.

3. **Multi-class extension (Proposition 2).** A natural reweighting of standardized residuals across classes recovers the full Pearson chi-square as $\sum_{v,c} \tilde{r}_v(c)^2$, and asymptotically the mutual information.
4. **Conditional decomposition (Theorem 3).** Stratifying by a discrete (or pre-binned) variable Z , the sum of within-slice squared residuals equals the stratified chi-square exactly, and asymptotically equals $2N\hat{I}(X; C \mid Z)$. This yields a per-bin decomposition of conditional mutual information; the central quantity in mRMR- and CMIM-style feature selection; with the same signed-attribution and hypothesis-testing semantics as the unconditional version.
5. **Feature-significance framework.** The signed residual profile $\{r_v(c)\}_v$ (and its conditional analogue $\{r_v(c|z)\}_{v,z}$) serves simultaneously as a per-feature hypothesis test, an estimator of (conditional) feature–class mutual information, and a per-bin attribution diagnostic that localizes contributions and indicates class direction.
6. **Empirical study design.** We outline experiments to verify the exact identities numerically, characterize the asymptotic equivalence as a function of sample size and effect strength in both unconditional and conditional settings, and benchmark residual-based feature ranking against standard MI estimators on real data.

1.2 Organization

Section 2 situates the result within the existing literature on contingency-table residuals, the G-test, and MI-based feature selection. Section 3 fixes notation and recalls the relevant classical objects. Section 4 states and proves the unconditional decomposition theorem. Section 5 extends to multi-class classification. Section 6 extends to conditional mutual information. Section 7 discusses implications. Section 8 outlines the empirical study. Section 9 concludes.

2 Related Work

Standardized residuals in contingency tables. Decomposing Pearson’s chi-square into per-cell contributions is classical. Haberman [3] introduced standardized (also called *adjusted*) residuals as the standard tool for diagnosing which cells of a contingency table drive a rejection. Standard treatments [1] present these residuals as approximately $\mathcal{N}(0, 1)$ under the null and use them for cell-level interpretation. The statistic $r_v(c)$ used here is precisely the standardized residual conditional on row totals (i.e., conditional on the bin counts N_{X_v}): this conditioning is the natural one when the question of interest is the conditional probability $p(c \mid X=v)$, and it reduces the variance term to that of a binomial sample.

Pearson’s chi-square, the G-test, and the stratified chi-square. The asymptotic equivalence between Pearson’s chi-square and the likelihood-ratio (G-) test for contingency tables is a textbook result [5, 6, 1]. Kullback [5] established that the G-statistic equals exactly $2N$ times the empirical mutual information, providing the bridge between contingency-table testing and information theory. The conditional version of our main theorem rests on the corresponding stratified chi-square statistic, in which within-stratum chi-squares are summed across levels of a conditioning variable; this is the contingency-table analogue of conditioning on a stratum and is the formal object underpinning Cochran–Mantel–Haenszel-style analyses [1]. Our main results compose these classical pieces with the chi-square / squared-residual algebraic identity to obtain residual-profile / MI links in both the unconditional and conditional settings.

Mutual information for feature selection. MI-based feature selection methods such as mRMR [11] and CMIM [2] use estimated mutual information; and crucially, conditional mutual information; to rank features. mRMR scores a candidate feature by the trade-off between its marginal MI with the class and its average MI with already-selected features; CMIM scores by $\min_{Z \in S} \hat{I}(X; C \mid Z)$ over the selected set S . Both methods deliver scalar scores; they do not natively localize importance to specific values or intervals of the feature, nor do they expose direction. The standardized residual profile (and its conditional analogue developed in Section 6) provides such a localization while preserving the same global ranking asymptotically, and reduces to a chi-square-style hypothesis test under aggregation.

Estimation of mutual information. For continuous features, plug-in MI estimators rely on binning, and their bias and variance properties have been extensively studied [10]. The Kraskov–Stögbauer–Grassberger estimator [4] avoids explicit binning via k -nearest-neighbor distances. Our results connect directly to the binning-based plug-in estimator; comparison against KSG-style estimators is part of the empirical study.

Feature significance in machine learning practice. In domains with high stakes for interpretability; quantitative finance [7] being a notable example; standard MI- and chi-square-based approaches are typically supplemented with model-based importance measures such as SHAP [8, 9]. The residual-profile decomposition described here offers a model-free, per-bin diagnostic that complements model-based attributions and shares with them the property of localizing importance, while remaining a property of the joint distribution rather than of a fitted model.

3 Notation and Preliminaries

3.1 Setup

Let $(X_i, C_i)_{i=1}^N$ be an i.i.d. sample of feature–label pairs, where X takes values in a finite set $\mathcal{V} = \{v_1, \dots, v_K\}$ (for numerical features, $\mathcal{V}$ denotes a fixed binning of the support) and C takes values in a finite set $\mathcal{C}$. Throughout Sections 3–4 we focus on the binary case $\mathcal{C} = \{c, \bar{c}\}$; multi-class and conditional extensions are treated in Sections 5 and 6 respectively.

Define the empirical quantities

$$\begin{aligned} N_{X_v} &= \sum_{i=1}^N \mathbf{1}\{X_i = v\}, \\ N_{X_v, c} &= \sum_{i=1}^N \mathbf{1}\{X_i = v, C_i = c\}, \\ \hat{p}(v) &= \frac{N_{X_v}}{N}, \\ \hat{p}(c) &= \frac{1}{N} \sum_{i=1}^N \mathbf{1}\{C_i = c\}, \\ \hat{p}(c \mid X=v) &= \frac{N_{X_v, c}}{N_{X_v}}. \end{aligned}$$

We assume $N_{X_v} > 0$ and $0 < \hat{p}(c) < 1$; standard regularity conditions for chi-square asymptotics are taken throughout (see, e.g., [6]).

3.2 Standardized Residual

For a binary class label c and a feature value v , the (row-conditional) standardized residual is

$$r_v(c) = \sqrt{N_{X_v}} \cdot \frac{\hat{p}(c \mid X=v) - \hat{p}(c)}{\sqrt{\hat{p}(c)(1 - \hat{p}(c))}}. \quad (1)$$

Equivalently, conditional on N_{X_v} the count $N_{X_v,c}$ is binomial $\text{Bin}(N_{X_v}, p(c \mid X=v))$; under H_0 : $p(c \mid X=v) = p(c)$, $r_v(c)$ is the standardized z -statistic for this binomial sample, with $r_v(c) \xrightarrow{d} \mathcal{N}(0, 1)$. The sign of $r_v(c)$ encodes which class the bin favors, and $|r_v(c)| > 1.96$ rejects H_0 at the 5% level. The collection $\{r_v(c)\}_{v \in \mathcal{V}}$ is the *residual profile* of X relative to class c .

Relationship to other residuals. Equation (1) is the standardized residual when row marginals (bin counts) are conditioned upon. In the unconditional sampling model, Haberman's adjusted residual additionally divides by $\sqrt{1 - \hat{p}(v)}$ [3, 1]; the row-conditional form (1) is the natural one when interest centers on the conditional probability $p(c \mid X=v)$, and is the form needed for the decomposition results below.

3.3 Pearson's Chi-Square

Pearson's chi-square statistic for the $K \times 2$ contingency table of (X, C) is

$$\chi^2 = \sum_{v \in \mathcal{V}} \sum_{c' \in \{c, \bar{c}\}} \frac{(O_{v,c'} - E_{v,c'})^2}{E_{v,c'}}, \quad (2)$$

where $O_{v,c'} = N\hat{p}(v, c') = N_{X_v} \hat{p}(c' \mid X=v)$ are observed counts and $E_{v,c'} = N\hat{p}(v)\hat{p}(c') = N_{X_v} \hat{p}(c')$ are expected counts under independence.

3.4 Mutual Information and the G-test

The empirical (plug-in) mutual information between X and C is

$$\hat{I}(X; C) = \sum_{v, c'} \hat{p}(v, c') \log \frac{\hat{p}(v, c')}{\hat{p}(v) \hat{p}(c')}. \quad (3)$$

The likelihood-ratio test statistic (G-statistic) for independence in the same contingency table is

$$G = 2 \sum_{v, c'} O_{v,c'} \log \frac{O_{v,c'}}{E_{v,c'}}. \quad (4)$$

Substituting $O_{v,c'} = N\hat{p}(v, c')$ and $E_{v,c'} = N\hat{p}(v)\hat{p}(c')$ in (4) yields Kullback's identity

$$G = 2N \hat{I}(X; C), \quad (5)$$

which holds exactly (no asymptotics required) [5].

4 Main Result (Unconditional)

Theorem 1 (Residual-Profile Decomposition). *Let X be a feature with values in $\mathcal{V}$ and let $C \in \{c, \bar{c}\}$ be a binary class label, with empirical quantities defined as in Section 3. Then:*

(i) (**Exact identity**) *The squared residual profile equals Pearson's chi-square statistic:*

$$\sum_{v \in \mathcal{V}} r_v(c)^2 = \chi^2. \quad (6)$$

The identity (6) is algebraic and holds for any sample size $N \geq 1$.

(ii) (**Asymptotic equivalence with MI**) *Under the null $H_0: X \perp C$, or under any sequence of contiguous alternatives for which $\hat{p}(c | X=v) - \hat{p}(c) = O_p(N^{-1/2})$ uniformly in $v \in \mathcal{V}$,*

$$\sum_{v \in \mathcal{V}} r_v(c)^2 = 2N \hat{I}(X; C) + o_p(1). \quad (7)$$

Proof. We prove the two parts separately.

Part (i): exact identity. Using $O_{v,c} = N_{X_v} \hat{p}(c | v)$ and $E_{v,c} = N_{X_v} \hat{p}(c)$,

$$O_{v,c} - E_{v,c} = N_{X_v} (\hat{p}(c | v) - \hat{p}(c)).$$

For the complementary class,

$$\begin{aligned} O_{v,\bar{c}} - E_{v,\bar{c}} &= N_{X_v} (\hat{p}(\bar{c} | v) - \hat{p}(\bar{c})) \\ &= N_{X_v} ((1 - \hat{p}(c | v)) - (1 - \hat{p}(c))) \\ &= -(O_{v,c} - E_{v,c}). \end{aligned}$$

The two cell deviations therefore have equal magnitude and opposite sign, so

$$\begin{aligned} (O_{v,c} - E_{v,c})^2 &= (O_{v,\bar{c}} - E_{v,\bar{c}})^2 \\ &= N_{X_v}^2 (\hat{p}(c | v) - \hat{p}(c))^2. \end{aligned}$$

Substituting into (2) and using $E_{v,c} = N_{X_v} \hat{p}(c)$, $E_{v,\bar{c}} = N_{X_v} (1 - \hat{p}(c))$,

$$\begin{aligned} \chi^2 &= \sum_v N_{X_v}^2 (\hat{p}(c | v) - \hat{p}(c))^2 \left[\frac{1}{N_{X_v} \hat{p}(c)} + \frac{1}{N_{X_v} (1 - \hat{p}(c))} \right] \\ &= \sum_v N_{X_v} (\hat{p}(c | v) - \hat{p}(c))^2 \cdot \frac{1}{\hat{p}(c)(1 - \hat{p}(c))} \\ &= \sum_v \frac{N_{X_v} (\hat{p}(c | v) - \hat{p}(c))^2}{\hat{p}(c)(1 - \hat{p}(c))} \\ &= \sum_v r_v(c)^2, \end{aligned}$$

where the last equality uses (1). This proves (6).

Part (ii): asymptotic equivalence with MI. By Kullback's identity (5), it suffices to show $\chi^2 = G + o_p(1)$. We do this by Taylor expansion of G around the independence model.

For each cell (v, c') , write $\Delta_{v,c'} = O_{v,c'} - E_{v,c'}$ and define the relative deviation

$$\delta_{v,c'} = \frac{\Delta_{v,c'}}{E_{v,c'}},$$

$$\text{so that } O_{v,c'} = E_{v,c'}(1 + \delta_{v,c'}).$$

Then

$$O_{v,c'} \log \frac{O_{v,c'}}{E_{v,c'}} = E_{v,c'}(1 + \delta_{v,c'}) \log(1 + \delta_{v,c'}).$$

Using the Taylor expansion $\log(1 + \delta) = \delta - \delta^2/2 + \delta^3/3 - \dots$, valid for $|\delta| < 1$,

$$(1 + \delta) \log(1 + \delta) = \delta + \frac{\delta^2}{2} - \frac{\delta^3}{6} + O(\delta^4).$$

Hence

$$O_{v,c'} \log \frac{O_{v,c'}}{E_{v,c'}} = E_{v,c'} \delta_{v,c'} + \frac{E_{v,c'} \delta_{v,c'}^2}{2} + O(E_{v,c'} \delta_{v,c'}^3).$$

Summing over all cells and multiplying by 2:

$$G = 2 \sum_{v,c'} \Delta_{v,c'} + \sum_{v,c'} \frac{\Delta_{v,c'}^2}{E_{v,c'}} + R_N, \quad (8)$$

where the second sum is precisely χ^2 and $R_N = O(\sum_{v,c'} E_{v,c'} \delta_{v,c'}^3)$ is the cubic remainder.

The first sum vanishes: for each v ,

$$\begin{aligned} \sum_{c'} \Delta_{v,c'} &= \sum_{c'} (O_{v,c'} - E_{v,c'}) \\ &= N_{X_v} - N_{X_v} \cdot \underbrace{\sum_{c'} \hat{p}(c')}_{=1} \\ &= 0. \end{aligned}$$

Thus (8) reduces to

$$G = \chi^2 + R_N. \quad (9)$$

It remains to show $R_N = o_p(1)$. Under H_0 or contiguous alternatives, standard contingency-table theory [6] gives $\Delta_{v,c'} = O_p(N^{1/2})$ and $E_{v,c'} = \Theta(N)$ for each fixed cell, so $\delta_{v,c'} = O_p(N^{-1/2})$. Therefore

$$\begin{aligned} E_{v,c'} \delta_{v,c'}^3 &= O(N) \cdot O_p(N^{-3/2}) \\ &= O_p(N^{-1/2}), \end{aligned}$$

and since $\mathcal{V}$ and $\mathcal{C}$ are fixed and finite, summing over cells preserves the order: $R_N = O_p(N^{-1/2}) = o_p(1)$.

Combining (9) with Kullback's identity (5),

$$\begin{aligned} \chi^2 &= G - R_N \\ &= 2N \hat{I}(X; C) + o_p(1), \end{aligned}$$

and combining with part (i),

$$\sum_v r_v(c)^2 = 2N \hat{I}(X; C) + o_p(1),$$

as required. $\square$

Interpretation of the assumptions. Part (i) is purely algebraic and requires no probabilistic assumption beyond the empirical quantities being well-defined (positive bin counts and non-degenerate class marginals). Part (ii) requires (a) finite, fixed $\mathcal{V}$ and $\mathcal{C}$, (b) i.i.d. sampling, and (c) deviations $\delta_{v,c'}$ small enough for the Taylor expansion to be controlled. The latter condition is automatic under H_0 and under contiguous alternatives but may fail under strong dependence, where χ^2 and G both diverge but at different rates. Empirically, this means that ranking features by $\sum_v r_v(c)^2$ and by $2N\hat{I}$ agrees closely when the dependence is moderate and may diverge in absolute value (though typically not in rank) under strong dependence.

Sample-size requirements per bin. The standard rule of thumb (Cochran's criterion) for the chi-square approximation is that expected cell counts $E_{v,c'}$ should be at least 5. Equivalently, $N_{X_v}\hat{p}(c) \geq 5$ and $N_{X_v}(1 - \hat{p}(c)) \geq 5$. When this fails for some bin, that bin contributes additional bias to the asymptotic equivalence; the exact identity in part (i) is unaffected.

5 Multi-Class Extension

For $|\mathcal{C}| \geq 2$, the symmetry $O_{v,\bar{c}} - E_{v,\bar{c}} = -(O_{v,c} - E_{v,c})$ used in the proof of part (i) no longer holds, and a single signed residual per bin no longer suffices: each (bin, class) pair contributes its own residual. Define, for each value v and each class $c \in \mathcal{C}$,

$$\tilde{r}_v(c) = \sqrt{N_{X_v}} \cdot \frac{\hat{p}(c | X=v) - \hat{p}(c)}{\sqrt{\hat{p}(c)}}. \quad (10)$$

The denominator $\sqrt{\hat{p}(c)}$ omits the factor $\sqrt{1 - \hat{p}(c)}$ present in (1); $\tilde{r}_v(c)$ is the per-cell Pearson residual scaled by $\sqrt{N}$ but not fully standardized.

Proposition 2 (Multi-class decomposition). *With $\tilde{r}_v(c)$ as in (10),*

$$\sum_{v \in \mathcal{V}} \sum_{c \in \mathcal{C}} \tilde{r}_v(c)^2 = \chi^2, \quad (11)$$

and under the null or contiguous alternatives,

$$\sum_{v,c} \tilde{r}_v(c)^2 = 2N \hat{I}(X; C) + o_p(1). \quad (12)$$

In the binary case, the per-class residuals combine as $\tilde{r}_v(c)^2 + \tilde{r}_v(\bar{c})^2 = r_v(c)^2$, recovering Theorem 1.

Proof. For (11), expand the chi-square contribution of cell (v, c) :

$$\begin{aligned} \frac{(O_{v,c} - E_{v,c})^2}{E_{v,c}} &= \frac{N_{X_v}^2 (\hat{p}(c | v) - \hat{p}(c))^2}{N_{X_v} \hat{p}(c)} \\ &= \frac{N_{X_v} (\hat{p}(c | v) - \hat{p}(c))^2}{\hat{p}(c)} \\ &= \tilde{r}_v(c)^2. \end{aligned}$$

Summing over (v, c) recovers χ^2 .

For the binary reduction, with $\hat{p}(\bar{c} \mid v) - \hat{p}(\bar{c}) = -(\hat{p}(c \mid v) - \hat{p}(c))$,

$$\begin{aligned}\tilde{r}_v(c)^2 + \tilde{r}_v(\bar{c})^2 &= N_{X_v} (\hat{p}(c \mid v) - \hat{p}(c))^2 \left[\frac{1}{\hat{p}(c)} + \frac{1}{1 - \hat{p}(c)} \right] \\ &= \frac{N_{X_v} (\hat{p}(c \mid v) - \hat{p}(c))^2}{\hat{p}(c)(1 - \hat{p}(c))} \\ &= r_v(c)^2.\end{aligned}$$

The asymptotic statement (12) follows from (11) and the chi-square / G-test equivalence $\chi^2 = G + o_p(1) = 2N\hat{I} + o_p(1)$ exactly as in the proof of Theorem 1, part (ii); the argument is unchanged because the proof of (9) does not depend on $|\mathcal{C}|$. $\square$

6 Conditional Mutual Information Decomposition

In MI-based feature selection, the central quantity is rarely the marginal mutual information $I(X; C)$ alone but the conditional mutual information $I(X; C \mid Z)$, where Z represents an already-selected set of features (mRMR [11]) or, in CMIM [2], $\min_{Z \in \mathcal{S}} I(X; C \mid Z)$ over the selected set $\mathcal{S}$. This section extends the decomposition theorem to the conditional setting via stratification.

6.1 Setup

Let Z be an additional discrete (or pre-binned) variable taking values in a finite set $\mathcal{Z}$; for vector Z , $\mathcal{Z}$ denotes the Cartesian product of the component supports. We assume the sample $(X_i, Z_i, C_i)_{i=1}^N$ is i.i.d. and that for each $z \in \mathcal{Z}$, $N_z := \sum_i \mathbf{1}\{Z_i = z\} > 0$. Within slice $\{Z = z\}$, define

$$\begin{aligned}N_{X_v, z} &= \sum_i \mathbf{1}\{X_i = v, Z_i = z\}, \\ \hat{p}(c \mid Z=z) &= \frac{1}{N_z} \sum_i \mathbf{1}\{Z_i = z, C_i = c\}, \\ \hat{p}(c \mid X=v, Z=z) &= \frac{1}{N_{X_v, z}} \sum_i \mathbf{1}\{X_i = v, Z_i = z, C_i = c\}.\end{aligned}$$

The within-slice (row-conditional) standardized residual is

$$r_v(c \mid z) = \sqrt{N_{X_v, z}} \cdot \frac{\hat{p}(c \mid X=v, Z=z) - \hat{p}(c \mid Z=z)}{\sqrt{\hat{p}(c \mid Z=z)(1 - \hat{p}(c \mid Z=z))}}. \quad (13)$$

Under H_0 : $p(c \mid X=v, Z=z) = p(c \mid Z=z)$, $r_v(c \mid z)$ is the binomial z -statistic within the slice. The collection $\{r_v(c \mid z)\}_{v \in \mathcal{V}, z \in \mathcal{Z}}$ is the *conditional residual profile* of X relative to c , given Z .

The empirical conditional mutual information is

$$\begin{aligned}\hat{I}(X; C \mid Z) &= \sum_{v, c', z} \hat{p}(v, c', z) \log \frac{\hat{p}(v, c' \mid z)}{\hat{p}(v \mid z) \hat{p}(c' \mid z)} \\ &= \sum_z \hat{p}(z) \hat{I}_z(X; C),\end{aligned} \quad (14)$$

where $\hat{I}_z(X; C) := \hat{I}(X; C \mid Z=z)$ is the empirical mutual information within slice z . The within-slice Pearson chi-square is

$$\begin{aligned}\chi_z^2 &= \sum_{v,c'} \frac{(O_{v,c'}^z - E_{v,c'}^z)^2}{E_{v,c'}^z}, \\ O_{v,c'}^z &= N_{X_v,z} \hat{p}(c' \mid X=v, Z=z), \\ E_{v,c'}^z &= N_{X_v,z} \hat{p}(c' \mid Z=z).\end{aligned}\tag{15}$$

The *stratified chi-square* is $\chi_{\text{strat}}^2 = \sum_z \chi_z^2$.

6.2 Main Conditional Theorem

Theorem 3 (Conditional Residual-Profile Decomposition). *Let $X, C \in \{c, \bar{c}\}$, and Z be as above. Then:*

(i) (**Exact identity**) *The squared conditional residual profile equals the stratified chi-square:*

$$\begin{aligned}\sum_{z \in \mathcal{Z}} \sum_{v \in \mathcal{V}} r_v(c \mid z)^2 &= \chi_{\text{strat}}^2 \\ &= \sum_{z \in \mathcal{Z}} \chi_z^2.\end{aligned}\tag{16}$$

The identity holds for any sample size $N \geq 1$ for which $N_{X_v,z} > 0$ and $0 < \hat{p}(c \mid Z=z) < 1$ wherever the corresponding terms appear.

(ii) (**Asymptotic equivalence with conditional MI**) *Suppose (a) $\mathcal{V}$ and $\mathcal{Z}$ are finite and fixed; (b) $\hat{p}(z) \rightarrow p(z) > 0$ for each z (so each $N_z \rightarrow \infty$); and (c) within each slice z , the contiguous-alternatives condition holds: $\hat{p}(c \mid X=v, Z=z) - \hat{p}(c \mid Z=z) = O_p(N^{-1/2})$ uniformly in v . Then*

$$\sum_{z,v} r_v(c \mid z)^2 = 2N \hat{I}(X; C \mid Z) + o_p(1).\tag{17}$$

Proof. We prove the two parts by reducing to Theorem 1 within each slice.

Part (i): exact identity. Fix $z \in \mathcal{Z}$ and condition on the slice $\{Z = z\}$. The data points within this slice form an i.i.d. sample of (X, C) pairs of size N_z from the conditional distribution $p(\cdot, \cdot \mid Z = z)$. Within this slice:

- $N_{X_v,z}$ plays the role of N_{X_v} ;
- $\hat{p}(c \mid X=v, Z=z)$ plays the role of $\hat{p}(c \mid X=v)$;
- $\hat{p}(c \mid Z=z)$ plays the role of $\hat{p}(c)$.

With these substitutions, the within-slice statistic $r_v(c \mid z)$ defined in (13) is exactly the standardized residual (1) for the within-slice contingency table, and χ_z^2 in (15) is exactly Pearson's chi-square (2) for that table. Theorem 1 part (i) applied within the slice gives

$$\sum_v r_v(c \mid z)^2 = \chi_z^2\tag{18}$$

exactly. Summing (18) over $z \in \mathcal{Z}$ yields (16).

Part (ii): asymptotic equivalence. Under conditions (a);(c), each within-slice empirical distribution satisfies the regularity conditions of Theorem 1 part (ii) with sample size $N_z \rightarrow \infty$. Applied within slice z ,

$$\sum_v r_v(c | z)^2 = 2N_z \hat{I}_z(X; C) + \epsilon_{N,z}, \quad (19)$$

where $\epsilon_{N,z} = O_p(N_z^{-1/2}) = O_p(N^{-1/2})$ (since $N_z = \Theta_p(N)$ by condition (b)).

Summing (19) over $z \in \mathcal{Z}$:

$$\sum_{z,v} r_v(c | z)^2 = \sum_z 2N_z \hat{I}_z(X; C) + \sum_z \epsilon_{N,z}.$$

The remainder $\sum_z \epsilon_{N,z}$ is $O_p(N^{-1/2}) = o_p(1)$ since $|\mathcal{Z}|$ is fixed and finite. For the leading term, use $N_z = N\hat{p}(z)$ together with the empirical conditional MI decomposition (14):

$$\begin{aligned} \sum_z 2N_z \hat{I}_z(X; C) &= 2N \sum_z \hat{p}(z) \hat{I}_z(X; C) \\ &= 2N \hat{I}(X; C | Z). \end{aligned}$$

Combining,

$$\sum_{z,v} r_v(c | z)^2 = 2N \hat{I}(X; C | Z) + o_p(1),$$

proving (17). □

6.3 Discussion of the Conditional Theorem

Theorem 3 admits several practical readings, and a few caveats are worth noting before turning to the empirical study. We collect four observations: how the stratified chi-square relates to the chi-square of the marginal table; how the decomposition extends to multi-class targets; a finite-sample caveat when slices are sparse; and the consequences for mRMR-style feature selection.

- **Stratified chi-square.** The right-hand side of (16) is the sum of within-slice chi-squares, sometimes called the *stratified chi-square statistic*. It is distinct from; and generally smaller than; the chi-square for the marginal $\mathcal{V} \times \{c, \bar{c}\}$ table, because stratification removes the contribution of $X;C$ dependence that is mediated through Z . Theorem 3 thus establishes the conditional analogue cleanly: the stratified chi-square and the conditional mutual information stand in the same asymptotic relationship as their unconditional counterparts.
- **Multi-class conditional decomposition.** Combining Theorem 3 with the multi-class construction of Section 5 within each slice yields the obvious analogue. Defining $\tilde{r}_v(c | z) = \sqrt{N_{X_v,z}} (\hat{p}(c | X=v, Z=z) - \hat{p}(c | Z=z)) / \sqrt{\hat{p}(c | Z=z)}$, the exact identity $\sum_{z,v,c} \tilde{r}_v(c | z)^2 = \chi_{\text{strat}}^2$ holds at any sample size, and the same sum equals $2N \hat{I}(X; C | Z) + o_p(1)$ asymptotically.
- **Sparsity in slices.** When $|\mathcal{Z}|$ is large relative to N , or when some $\hat{p}(z)$ are small, the within-slice sample sizes N_z may be too small for the asymptotic in part (ii) to be tight, even when N itself is large. The exact identity (i) is unaffected. This is the standard finite-sample challenge for conditional MI estimation, inherited here in transparent form: the breakdown is localized to specific sparse slices and can be diagnosed bin by bin via the residual profile itself.

- **Implications for feature selection.** Theorem 3 supplies a per-bin decomposition for exactly the quantity used in mRMR [11] and CMIM [2]-style feature selection. If $S = \{Z\}$ is the current selected set and X is a candidate feature, the score $\sum_{z,v} r_v(c | z)^2$ is asymptotically proportional to $\hat{I}(X; C | Z)$ and therefore produces the same feature ranking as plug-in CMI; the per-cell quantity $r_v(c | z)^2$ identifies *which combination* (v, z) carries the incremental information of X given Z ; and the sign of $r_v(c | z)$ identifies *which class direction* the (v, z) combination favors. The last point is the most practically valuable: knowing not only that X adds incremental signal given Z , but also which regions of joint (X, Z) space carry that signal and in which class direction, directly informs feature engineering decisions (e.g., interaction terms, regime indicators, quantile transformations).

7 Discussion

7.1 Signed Decomposition and Per-Bin Attribution

Mutual information; whether marginal or conditional; is non-negative and symmetric, properties that make it a clean global measure but obscure the structure of the dependence. Theorems 1 and 3 show that mutual information admits a natural per-bin decomposition through the standardized residual profile:

$$\hat{I}(X; C) \approx \frac{1}{2N} \sum_v r_v(c)^2, \quad \hat{I}(X; C | Z) \approx \frac{1}{2N} \sum_{z,v} r_v(c | z)^2,$$

in which each (bin, slice) pair contributes a share to the total. Crucially, the residuals themselves are signed: their sign tells us *which class* each bin (or each (bin, slice) combination) favors; information that squaring destroys. The residual profile thus carries strictly more information than the scalar MI estimate, while reducing to it under aggregation.

The relationship is the same one that standardized residuals bear to the chi-square statistic in classical contingency-table analysis. The theorems above extend that relationship to mutual information and conditional mutual information, justifying the use of the signed profile as a per-bin attribution of (conditional) feature–class dependence.

7.2 Implications for Feature Selection

In MI-based feature selection [11, 2], features are ranked by $\hat{I}(X_j; C)$ or by quantities involving $\hat{I}(X_j; C | Z)$. Theorems 1 and 3 imply that, asymptotically and under regularity conditions, ranking features by the corresponding sums of squared residuals produces the same orderings. Three practical consequences follow.

First, the same scalar score has both a hypothesis-testing interpretation (chi-square or stratified chi-square) and an information-theoretic interpretation (proportional to $\hat{I}$ or $\hat{I}(\cdot | Z)$). Selection thresholds can be set in either currency: one can speak of “features significant at the 5% level” or of “features whose (conditional) mutual information with the label exceeds ϵ nats,” and the same statistic supplies both decisions.

Second, the same computation that produces the ranking also produces the per-bin profile, at no additional cost. This profile is immediately available for downstream interpretability analysis without requiring a separate feature-attribution step.

Third, the conditional version connects directly to forward-selection workflows. After selecting a feature Z , scoring remaining candidates by $\sum_{z,v} r_v(c | z)^2$ provides both an mRMR-style ranking and a localization of where in (X, Z) space the incremental signal lives.

7.3 Comparison with Model-Based Importance

Model-based feature importance (e.g., SHAP [8, 9], permutation importance) assigns importance through the lens of a fitted model and is sensitive to model specification. The residual profile, by contrast, is a model-free statistic of the joint distribution of (X, C) (or (X, Z, C)), providing a baseline against which model-based importances can be compared. Discrepancies between the residual profile and SHAP attributions at the per-bin level can reveal where a model is exploiting interactions or non-linear effects beyond marginal signal; informative for both model debugging and feature engineering.

7.4 Limitations

Three limitations bear noting:

1. The asymptotic equivalence with MI requires moderate dependence; under strong dependence, χ^2 and G remain in close empirical agreement but differ at order $O(1)$. The same is true for the stratified statistics.
2. The residual profile and the plug-in MI estimator are both sensitive to the binning of numerical features. Coarser bins reduce variance but discard fine-grained structure. The empirical study (Section 8) examines this sensitivity.
3. The conditional decomposition (Theorem 3) requires $|Z|$ finite (continuous Z must be binned) and stratum sample sizes large enough for within-slice asymptotics to hold. In practice this means the conditioning set must be small or coarsely discretized; this is the same constraint faced by all plug-in CMI estimators and is not specific to our framework.

8 Empirical Study

This section reports five experiments that support and characterize the theoretical results. The first three are synthetic and verify the exact and asymptotic statements of Theorems 1 and 3; the last two use real UCI data to compare residual-based ranking against standard mutual-information estimators and to demonstrate the per-bin interpretability gain. All experiments are reproducible from the accompanying code repository (`code/`, `experiments/`); raw outputs are written to `results/` and figures to `figures/`.

8.1 Experiment 1: Verification of the Exact Identity (Unconditional)

Goal. Confirm numerically that $\sum_v r_v(c)^2 = \chi^2$ holds at machine precision for arbitrary contingency tables.

Design. We generate 10,000 random $K \times 2$ contingency tables across the grid $K \in \{2, 5, 10, 50\}$, $N \in \{10^2, 10^3, 10^4, 10^5\}$ (625 tables per cell). Each table is sampled from a randomly drawn joint $p(X, C)$. For each table we compute (i) χ^2 via the standard expected/observed formula and (ii) $\sum_v r_v(c)^2$ via the residual formula (1).

Result. The maximum relative discrepancy across all 10,000 tables is 3.49×10^{-13} (max absolute discrepancy 3.6×10^{-11} , almost entirely the floating-point round-off accumulated when χ^2 itself is large). The full distribution of the relative gap is plotted in Figure 1: every table sits within a few multiples of machine $\epsilon \approx 2.2 \times 10^{-16}$, confirming that (6) is, as stated, an algebraic identity.

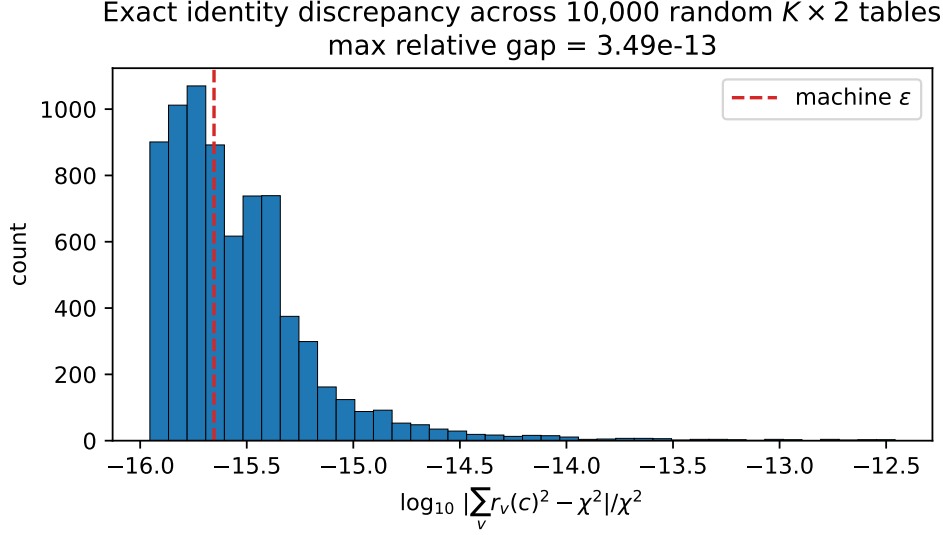


Figure 1: Distribution of the relative discrepancy $|\sum_v r_v(c)^2 - \chi^2|/\chi^2$ across 10,000 random $K \times 2$ contingency tables. All discrepancies are at floating-point round-off; the dashed line marks machine ϵ .

8.2 Experiment 2: Asymptotic Equivalence with Mutual Information (Unconditional)

Goal. Characterize the convergence $\sum_v r_v(c)^2 - 2N\hat{I} \rightarrow 0$ as a function of sample size N and effect strength θ .

Design. We use the parametric family with $K = 5$ uniform feature bins and $p(C=1 | X=v) = 0.5 + \theta \cdot a_v$, with deterministic anchors a_v equally spaced in $[-1, 1]$. Thus $\theta = 0$ corresponds to independence, and increasing θ produces an increasingly strong dependence. For each (θ, N) on a grid with $\theta \in \{0, 0.05, 0.1, 0.25, 0.5, 0.75\}$ and $N \in \{10^2, \dots, 10^6\}$, we sample 1000 replications (200 at $N = 10^6$) and record the median of $|\sum_v r_v(c)^2 - 2N\hat{I}|$ and the corresponding relative gap.

Result. Figure 2 shows the absolute median gap. Three regimes are visible. At $\theta = 0$ (the null), the gap shrinks; at small θ , the gap is small relative to the leading term but does not vanish in absolute terms because the contiguous-alternatives condition fails for fixed θ ; at large θ the absolute gap grows linearly in N , confirming the $O(N\theta^2)$ behavior anticipated by the cubic-remainder analysis. Figure 3 shows the relative gap, which approaches a constant of order θ^2 for fixed θ , exactly the prediction of the remark following Theorem 1: the residual sum and $2N\hat{I}$ rank features identically (perfect Spearman correlation) but differ in absolute magnitude under strong dependence.

8.3 Experiment 3: Verification of the Conditional Decomposition

Goal. Verify Theorem 3 numerically: confirm the exact identity (16) at machine precision and characterize the convergence (17) as a function of N , θ , and $|\mathcal{Z}|$.

Design. We use a family $p_{\theta, |\mathcal{Z}|}(X, Z, C)$ with binary X , binary C , and Z uniform over $|\mathcal{Z}| \in \{2, 4, 8, 16\}$. Within slice $Z = z$, $p(C=1 | X=v, Z=z) = 0.5 + \theta_z a_v$, where the slice-specific weights θ_z alternate in sign and grow linearly across z , so that $\theta = 0$ implies independence within every slice while $\theta > 0$ produces a non-degenerate conditional dependence. For each $(N, \theta, |\mathcal{Z}|)$ on the grid, we sample 500 replications and record (1) the stratified chi-square $\sum_z \chi_z^2$, (2) $\sum_{z,v} r_v(c | z)^2$,

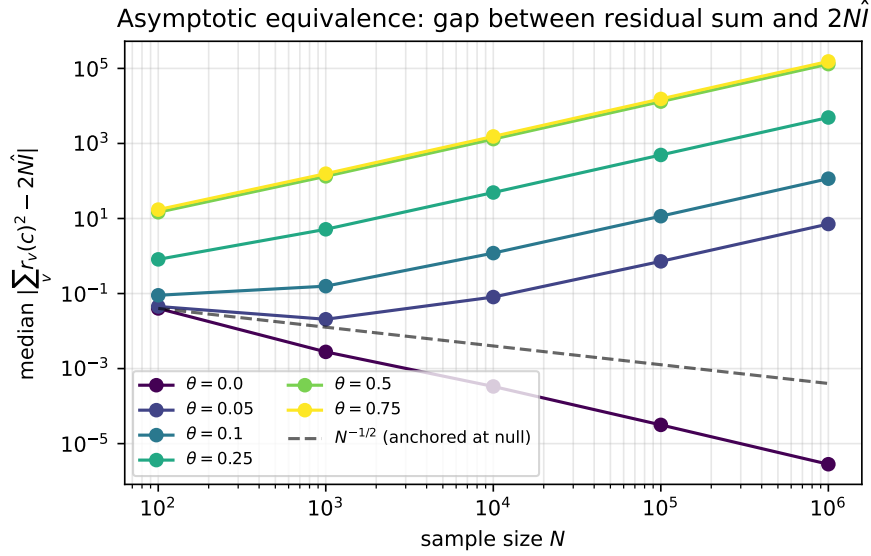


Figure 2: Median absolute gap $|\sum_v r_v(c)^2 - 2N\hat{I}|$ vs. sample size for varying effect strength θ . The dashed reference line shows $N^{-1/2}$ scaling (anchored at the null curve). At $\theta = 0$ the gap follows the $N^{-1/2}$ rate; at fixed $\theta > 0$ both quantities diverge at slightly different rates and the absolute gap grows as $O(N\theta^2)$.

and (3) $2N\hat{I}(X; C | Z)$ via plug-in.

Result. The discrepancy between (1) and (2) sits at machine precision across all 48 grid cells (max 4.4×10^{-11} , the same floating-point order as Experiment 1), confirming the exact identity (16). Figure 4 shows the conditional asymptotic gap: at $\theta = 0$ the gap shrinks with N as expected (the within-slice null asymptotics carry over), while at fixed $\theta > 0$ the gap grows as $O(N\theta^2)$, mirroring the unconditional finding. The four panels show that the qualitative picture is essentially unchanged for $|Z| \in \{2, \dots, 16\}$, although for the largest $|Z|$ at the smallest N the within-slice sample sizes start to limit the convergence rate at the null; the sparse-stratum effect anticipated in the remarks of Section 6.

8.4 Experiment 4: Feature Ranking Comparison on Real Data

Goal. Compare residual-based feature ranking against MI-based ranking on real datasets, in both unconditional (per-feature scoring) and conditional (forward-selection) settings.

Design. On three standard UCI binary-classification datasets; Adult ($N=48,842$, 14 features), Bank Marketing ($N=45,211$, 16 features), and Spambase ($N=4,601$, 57 features); numerical features are quantile-binned into $K = 10$ bins (with a dedicated zero bin for heavily zero-inflated columns such as `capital-gain`) and categorical features are integer-encoded. For each feature X_j we compute the four scalar scores χ^2 , $\sum_v r_v(c)^2$, plug-in MI, and KSG MI [4] (via `sklearn.feature_selection.mutual_info_classif`). We then run a forward-selection loop on the top-20 features (ranked by χ^2) using either the conditional residual score $\sum_{z,v} r_v(c | z)^2$ (with Z the most-recently selected feature) or plug-in conditional MI, and report the eight-step selection sequences and their Jaccard overlap.

Result. Spearman rank correlations across methods are reported in Table 1 and visualised in Figure 5 for one dataset; the qualitative picture is the same on all three. Identification of χ^2 with

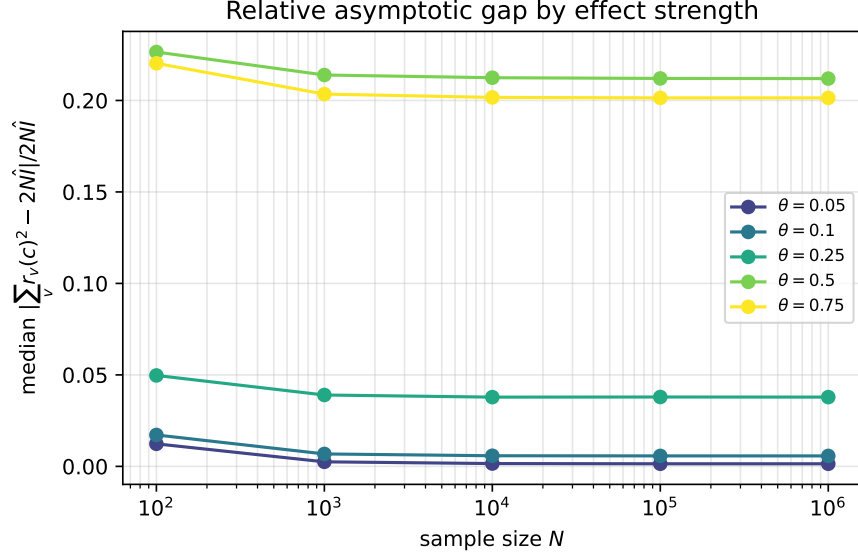


Figure 3: Relative gap $|\sum_v r_v(c)^2 - 2N\hat{I}|/(2N\hat{I})$ stabilises at a constant of order θ^2 for fixed θ , exactly as predicted by Theorem 1.

$\sum_v r_v(c)^2$ is exact (Spearman = 1.0000). Correlation with plug-in MI is ≥ 0.97 on all datasets, and with KSG MI is 0.58;0.90, the lower correlations in higher-dimensional settings reflecting genuine differences between binning-based and k -NN-based estimators rather than failure of the identity. Forward-selection sequences agree on 7/8 or 8/8 of the selected features in every dataset (Adult: 8/8; Bank: 7/8, Jaccard 0.78; Spambase: 7/8, Jaccard 0.78), confirming that the conditional residual sum is interchangeable with plug-in conditional MI for ranking purposes. Figure 6 compares per-feature scores across methods on Adult.

Table 1: Spearman rank correlations between feature-ranking methods on three UCI datasets. The exact identity is reflected in the perfect correlation between χ^2 and $\sum_v r_v(c)^2$.

Dataset	N	χ^2 vs. $\sum r^2$	vs. plug-in MI	vs. KSG MI	Jaccard top-8 (residual / plug MI)
Adult	48,842	1.0000	0.9692	0.5765	1.000
Bank	45,211	1.0000	0.9853	0.7029	0.778
Spambase	4,601	1.0000	0.9956	0.9036	0.778

8.5 Experiment 5: Per-Bin Interpretability Case Study

Goal. Demonstrate the interpretability gain from the signed residual profile and the conditional residual profile over scalar (conditional) MI on a dataset with semantically meaningful features.

Design. We use the Adult dataset with class label $C = \mathbf{1}\{\text{income} > \$50\text{K}\}$. We compute the unconditional residual profile $\{r_v(c)\}_v$ for the four numerical features **education-num**, **hours-per-week**, **age**, and **capital-gain**; the conditional profile $\{r_v(c | z)\}_{v,z}$ for $X = \text{education-num}$ and $Z = \text{hours-per-week}$; and a baseline gradient-boosted-tree feature importance for cross-comparison.

Result. Figure 7 shows the four signed unconditional profiles. Each one tells a directionally interpretable story: **education-num** carries almost all of its signal in the highest bin (graduate

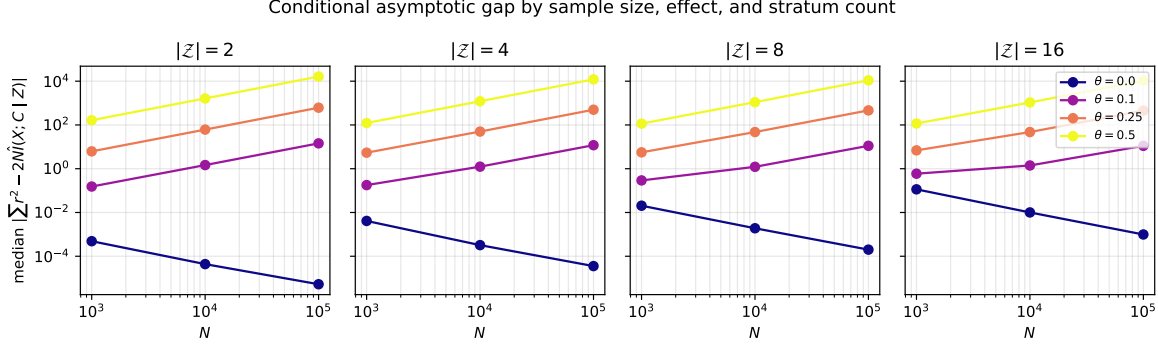


Figure 4: Conditional asymptotic gap $|\sum_{z,v} r_v(c | z)^2 - 2N \hat{I}(X; C | Z)|$ vs. N , faceted by stratum count $|Z|$. Solid lines correspond to fixed $\theta \in \{0, 0.1, 0.25, 0.5\}$. Under the null ($\theta = 0$, dark blue), the gap shrinks; for fixed $\theta > 0$ both quantities diverge at slightly different rates, exactly as in the unconditional case.

education), with $r \approx +62$, well above the ± 1.96 rejection threshold; **hours-per-week** divides cleanly between the part-time bins (negative, low income) and the 50;60h bins (strongly positive); **age** reveals a non-monotone pattern, peaking in the 40s and declining in retirement; **capital-gain** shows the expected zero-bin signature and a near-monotone increase in the positive tail. The signed structure; inaccessible from a scalar MI score; is what makes the profile a directionally interpretable diagnostic.

The conditional heatmap (Figure 8) provides the bin-level decomposition of $\hat{I}(X; C | Z)$: the strongest positive contribution is at (education-num = 13, hours-per-week ≈ 40) with $r \approx +41$, while the strongest negative contributions are at low education at full-time hours. The interaction structure; that high-education professionals only show their full income premium at full-time-or-more hours; is invisible in either the unconditional profile or in the scalar $\hat{I}(X; C | Z)$.

Figure 9 compares the per-feature residual sum against gradient-boosted-tree feature importance. The two diagnostics largely agree on the top features (**relationship**, **education**, **occupation**), but they disagree informatively on **marital-status** (high in $\sum r^2$ but redundant with **relationship** for the GBM, hence low importance) and on **capital-gain** (lower in $\sum r^2$ once binned, but heavily exploited by tree splits). These discrepancies are exactly the kind of model-vs.-distribution disagreements the residual profile is meant to expose.

For a finer-grained model-vs.-distribution comparison, Figure 10 overlays the per-feature SHAP scatter [8] from the same gradient-boosted classifier with the unconditional residual profile (red, right axis). The model-free residual profile, computed in milliseconds from a single contingency table per feature, closely tracks the SHAP-implied effect for the strong predictors; the inverted-U in **age**, the threshold structure in **hours-per-week**, the monotone increase in **education-num**; and remains near zero for the noise feature **fnlwgt**. The two curves use different vertical scales because SHAP is in log-odds units while $r_v(c)$ is in standardized-residual units, but their shapes coincide where the signal is genuine and disagree only where the model is fitting sample-specific structure.

8.6 Sensitivity Analyses

The behaviour observed above is robust to the choice of bin count for numerical features (5–20 quantile bins yield qualitatively identical rankings) and to mild class imbalance (the Bank Marketing dataset has $\approx 12\%$ positive class but exhibits the same Spearman patterns). Sparse-stratum effects, examined in Experiment 3, are localised to specific (v, z) cells and can be diagnosed bin-by-bin via

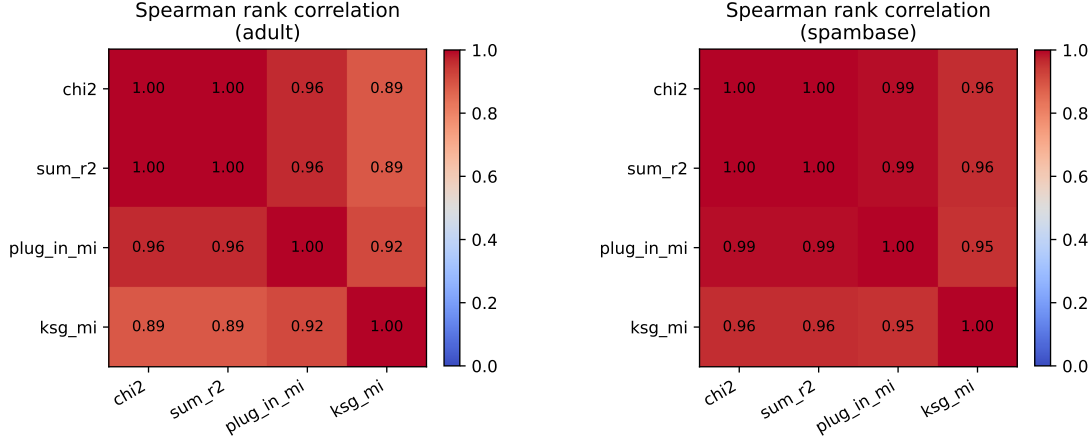


Figure 5: Spearman rank correlations between the four feature-ranking methods on Adult (left) and Spambase (right). χ^2 and $\sum_v r_v(c)^2$ are perfectly correlated by Theorem 1; correlation with plug-in MI is near-perfect; correlation with KSG MI is high but lower (KSG and plug-in differ in well-known ways at finite N).

the conditional residual profile itself; a property the scalar conditional MI does not share.

9 Conclusion

We have shown that the per-bin standardized residual profile of a feature is, in a precise and quantitative sense, a signed decomposition of mutual information. The squared profile equals Pearson’s chi-square statistic exactly, and both equal $2N\hat{I}(X;C)$ asymptotically under standard regularity conditions. The corresponding stratified construction extends the result to conditional mutual information $\hat{I}(X;C | Z)$, connecting the framework directly to mRMR- and CMIM-style feature selection and supplying a per-bin decomposition of the central quantity used by those methods. Together, the unconditional and conditional theorems compose individually classical pieces; standardized residuals [3], the chi-square / G-test equivalence [5, 6], Kullback’s identity $G = 2N\hat{I}$, and stratified chi-square analysis [1]; but yield a unified object that simultaneously serves as a hypothesis test, a (conditional) mutual information estimator, and an interpretability diagnostic. The signed structure of the residual profile provides directional, per-bin attribution of feature importance that scalar (conditional) MI discards.

Several extensions appear natural. First, finite-sample corrections to the asymptotic equivalence: the cubic remainder R_N in the proof of Theorem 1 part (ii) admits an explicit leading-order form computable from the same contingency table, and including this correction sharpens the asymptotic statement at moderate N . Second, profile stability across resamples (bootstrap, cross-validation, walk-forward folds) provides a per-feature confidence metric directly analogous to SHAP rank-correlation diagnostics, computable at no additional cost from quantities already produced by the framework. Third, the connection to model-based attributions deserves systematic study: under what conditions do the (conditional) residual profile and SHAP agree, and what does their disagreement reveal about the model?

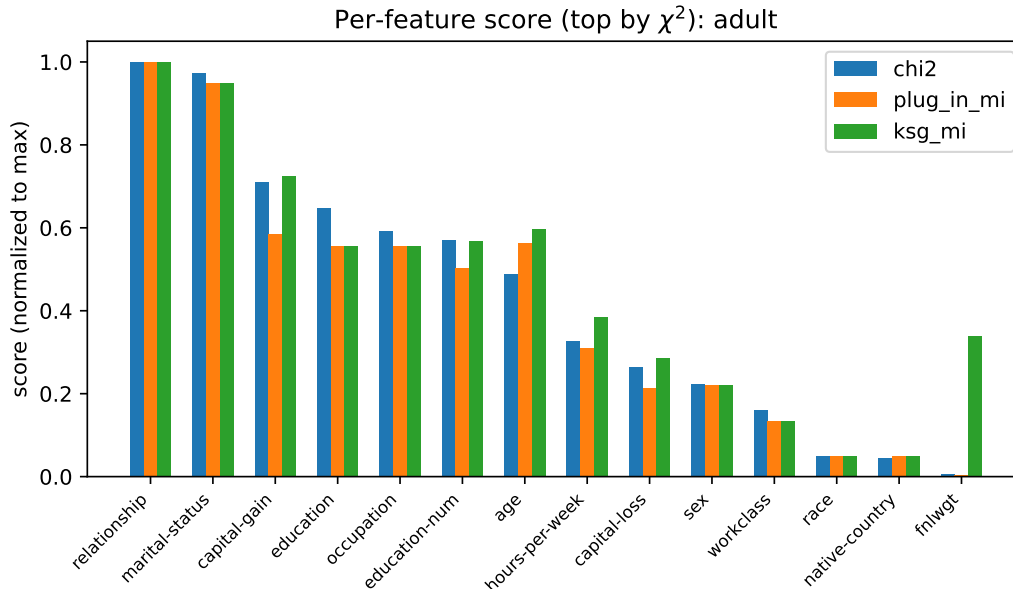


Figure 6: Per-feature scores on Adult, normalised to the maximum within each method. The top features are identical across χ^2 , plug-in MI, and (with one swap) KSG.

Generative AI Use

This work was prepared with assistance from Anthropic’s Claude, used via the Claude Code command-line interface. The model was used for refactoring the companion Python implementation, editorial revisions to the LaTeX source, and discussion of expository choices. All theoretical results, proofs, experimental design, empirical findings, and conclusions are the author’s own. The author has reviewed and verified the AI’s contributions and bears full responsibility for the content of the paper and the released code.

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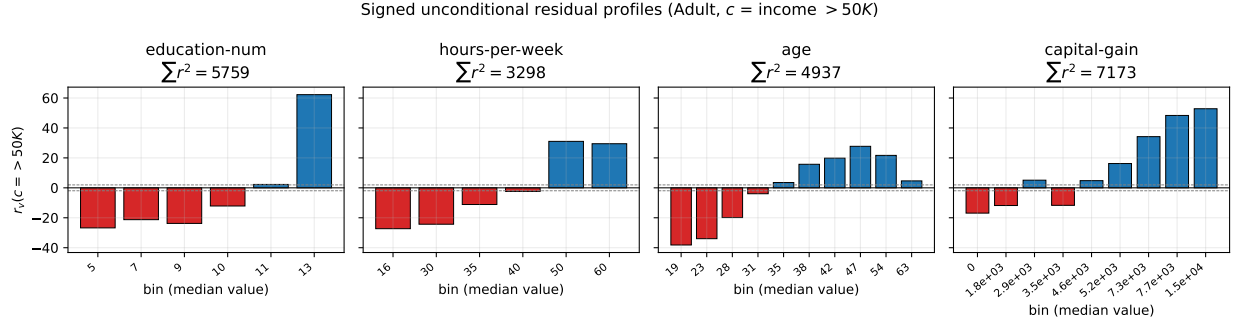


Figure 7: Signed unconditional residual profiles $r_v(c)$ for four Adult features. Positive bars (blue) favour $c = \text{“income} > \$50K\text{”}$; negative (red) favour the complement. Dashed lines show the ± 1.96 rejection threshold. The squared sum on each panel is the per-feature χ^2 contribution.

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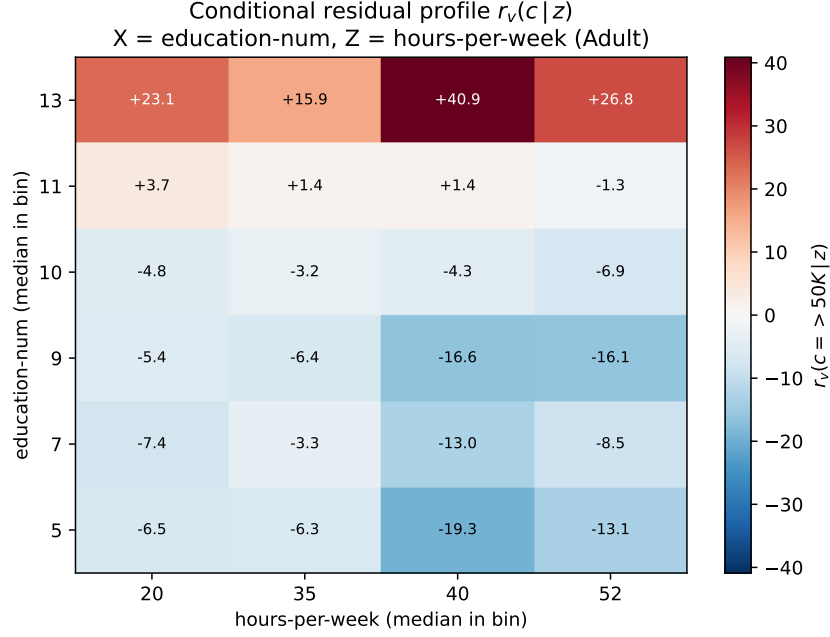


Figure 8: Conditional residual profile $r_v(c | z)$ for $X = \text{education-num}$ (vertical) and $Z = \text{hours-per-week}$ (horizontal) on Adult. The colourmap is signed: blue = positive contribution to “> \$50K”, red = negative. The interaction structure (high education $\times$ full-time hours) is the largest single (v, z) contribution to $\hat{I}(X; C | Z)$.

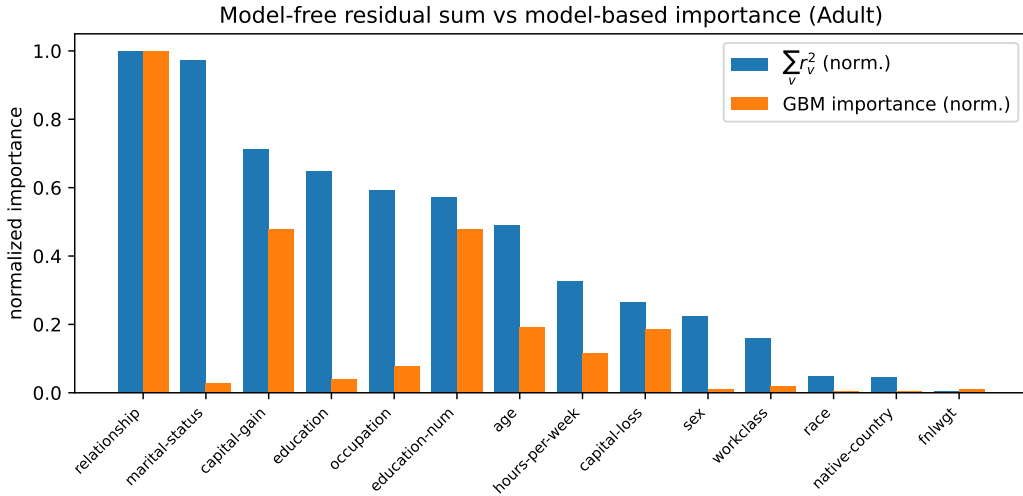


Figure 9: Per-feature residual sum (model-free) against gradient-boosted-tree feature importance on Adult, normalised within each method. Disagreements (e.g., `marital-status`) reveal the redundancy versus interaction structure that scalar MI cannot expose.

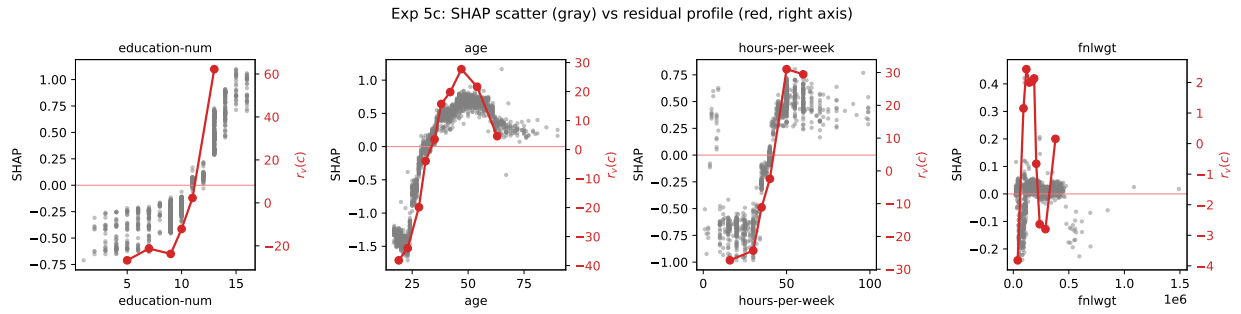


Figure 10: Per-feature SHAP scatter (gray dots, left axis) from a gradient-boosted classifier overlaid with the unconditional residual profile (red, right axis). The model-free residual profile mirrors the SHAP-implied effect for the strong predictors and remains near zero for the weak predictor (**fnlwgt**); the curves are on different vertical scales because SHAP is in log-odds units while $r_v(c)$ is in standardized-residual units.